AMITY UNIVERSITY

AMITY SCHOOL OF ENGINEERING & TECHNOLOGY

Amity University Uttar Pradesh, Greater Noida

MINOR PROJECT REPORT

(Course Code: ETMN100 — 4.00 Credit Units)

WaterTriage: A Data-Driven Water Quality Risk Scoring and Intervention Prioritization System for Uttar Pradesh and Bihar

Submitted in partial fulfilment of the requirements for the degree of

Bachelor of Technology in Computer Science & Engineering

Submitted by

Anubhav Anand

Enrollment No.: A41105223039 | Batch: 2023–2027 (Semester VII)

Under the Supervision of

Dr. Girish Paliwal

Faculty Guide, Department of Computer Science & Engineering

Academic Session 2026–27 (October 2026)

Candidate's Declaration

I, Anubhav Anand (Enrollment No. A41105223039), student of B.Tech (Computer Science & Engineering), Batch 2023–2027, Amity School of Engineering & Technology, Amity University Uttar Pradesh, Greater Noida, hereby declare that the Minor Project report titled "WaterTriage: A Data-Driven Water Quality Risk Scoring and Intervention Prioritization System for Uttar Pradesh and Bihar" represents original engineering and analytical work carried out by me under the supervision of Dr. Girish Paliwal during the period 20 July 2026 to 14 October 2026. All external datasets, standards, and scholarly sources are explicitly cited, and this work has not been submitted in part or in full to any other university or institution for the award of any degree or diploma.

Signature of Student: _____ Date: _____

Name: Anubhav Anand (Enrollment No. A41105223039)

Certificate of the Faculty Guide

This is to certify that the Minor Project (ETMN100) report titled "WaterTriage: A Data-Driven Water Quality Risk Scoring and Intervention Prioritization System for Uttar Pradesh and Bihar" submitted by Anubhav Anand (Enrollment No. A41105223039) is a bona fide record of independent work carried out by him under my supervision and guidance at the Amity School of Engineering & Technology, Amity University Uttar Pradesh, Greater Noida, in partial fulfilment of the requirements for the degree of Bachelor of Technology in Computer Science & Engineering.

Signature of Faculty Guide: _____ Date: _____

Dr. Girish Paliwal

Department of Computer Science & Engineering, ASET, Amity University Uttar Pradesh

Certificate of Originality & Plagiarism Verification (G-32 Regulations)

In compliance with the Non-Teaching Credit Courses (NTCC) Guidelines (Regulation G-32) of Amity University Uttar Pradesh, this Minor Project report has been screened for text similarity using the institutional Turnitin anti-plagiarism software prior to final evaluation. The verified similarity index is within the permissible institutional threshold (<= 15%, excluding bibliographic references, statutory standard tables, and front-matter declarations).

Verification Parameter	Recorded Institutional Detail
Candidate & Course	Anubhav Anand (A41105223039) | B.Tech CSE | ETMN100 Minor Project (4.00 Credits)
Screening Software & G-32 Cap	Turnitin Institutional Originality Check (Permissible Threshold: <= 15%)
Turnitin Submission ID & Date	_____ / October 2026
Verified Similarity Index (%)	_____ % (Verified Compliant under G-32: Yes / No)
Signatures (Student & Guide)	Student: _____ Faculty Guide (Dr. Girish Paliwal): _____

Acknowledgements

I express my sincere gratitude to my Faculty Guide, Dr. Girish Paliwal, Department of Computer Science & Engineering, Amity School of Engineering & Technology, Amity University Uttar Pradesh, Greater Noida, for his rigorous technical guidance, weekly reviews across all twelve WPR milestones, and constructive critique on statistical sensitivity auditing and multi-tier data validation. I also thank the Head of Department and faculty members of ASET for providing the computational environment and academic framework required to complete this study.

Table of Contents

Report Section / Chapter	Location
Front Matter (Declarations, Faculty Certificate, Turnitin Verification, Acknowledgements)	Pages 2–3
Abstract & Keywords	Page 4
1. Introduction (Problem Context, Surveillance Gap & Synopsis Objectives)	Page 5
2. Literature Review and Related Work (WQMIS, Gangetic Hydrogeochemistry & WQI Flaws)	Page 6
3. System Architecture and Methodology (Schema, BIS 10500 Engine, Lab-ID Recovery & 4-Tier Fusion)	Page 7
4. Empirical Results and Multi-Source Validation (3-Way Sensitivity & 113-District Triage)	Page 11
5. Discussion, REST API & 6-Tab React Dashboard Implementation	Page 18
6. Methodological Limitations	Page 21
7. Conclusion and Future Work	Page 22
References (BIS IS 10500:2012, Government Portals & Peer-Reviewed Literature)	Page 23
Appendix A: Formal Revision Note on Weekly Progress Reports (WPR-6 to WPR-12)	Page 24
Appendix B: Verified Claims & Codebase Traceability Ledger (origin/main @ 95b2162)	Page 26

Abstract

Rural drinking water security across the Indo-Gangetic alluvial plains of Uttar Pradesh (75 districts) and Bihar (38 districts) is persistently challenged by geogenic contaminants—specifically arsenic, fluoride, and dissolved iron—alongside anthropogenic nitrate and fecal coliform pollution. Although the Government of India's Jal Jeevan Mission (JJM) collects habitation-level laboratory and field testing records within the Water Quality Management Information System (WQMIS), its public reporting interfaces are structured for macroeconomic administrative tracking rather than source-level or district-level remediation triage. This project designs, implements, and audits WaterTriage, a reproducible, containerized decision-support system built with Python, FastAPI, SQLAlchemy 2.0, PostgreSQL 16, and a six-tab React 18 frontend that ingests public JJM water quality datasets, scores contamination severity against Bureau of Indian Standards (BIS) IS 10500:2012 limits, and exposes ranked intervention priority queues.

The system is evaluated across two complementary operational layers at repository commit 95b2162. First, on a frozen canonical corpus of 1,741 individual WQMIS laboratory test reports (Uttar Pradesh: 966; Bihar: 775; collected 2022-07-01 to 2022-11-19 across 6,792 discrete parameter readings), a three-way sensitivity audit compares the baseline tested-weight formulation (Model v1 at the IS 10500:2012 iron limit of 0.3 mg/L: 89 Critical, 392 High, 28 Medium, 1,232 Low; Bihar mean 18.36 vs. UP 12.68 despite medians of 0.00 vs. 4.08), a laboratory-limit sensitivity run (at the 1.0 mg/L iron threshold printed on all 876 raw WQMIS iron records: 5 Critical, 139 High, 30 Medium, 1,567 Low; UP mean 9.99 > Bihar 1.58), and a coverage-adjusted total-weight formulation (Model v2 normalized over TOTAL_BIS_WEIGHT = 6.0 across all 10 BIS parameters with 40/20/10 bands: 3 Critical, 54 High, 125 Medium, 1,559 Low; UP mean 4.72 > Bihar 2.32). This audit proves that Bihar's elevated baseline mean was an artifact of 70 single-parameter iron tests in Madhepura district (0.318–0.478 mg/L). Second, a non-destructive enrichment and four-tier fusion engine resolves 302 of 354 previously unassigned laboratory districts (85.3%) across 20 single-district laboratories via a two-step ASP.NET session cookie handshake (leaving only 52 samples [2.99%] in multi-district regional lab L158 and 0% missing in Bihar), recovers 305 village/habitation names (17.52%, up from 1.21%), and integrates official Form WQ6 village census tables (113 districts), the deduplicated 2009–2012 IMIS registry (100,690 events across all 93/93 historical districts), and CGWB 2012–2021 station series to achieve 100% risk-scoring coverage across all 113 canonical districts (46 Critical [32 pathogen overrides, 13 arsenic escalations, 1 score-driven], 27 High, 19 Medium, and 21 Low).

Keywords: Water Quality Surveillance; Jal Jeevan Mission (JJM); WQMIS; BIS IS 10500:2012; Multi-Tier Risk Scoring; Panel-Size Bias; FastAPI; React Decision-Support System.

1. Introduction

Shallow and intermediate alluvial aquifers along the Middle Gangetic Plain supply drinking water to over three hundred million residents across Uttar Pradesh and Bihar. Because these unconsolidated Holocene and Pleistocene sediments undergo reductive dissolution of iron oxyhydroxides and localized weathering of fluoride-bearing minerals, rural groundwater sources frequently exceed safe drinking water thresholds for arsenic, fluoride, and iron, while shallow handpumps remain vulnerable to nitrate leaching and bacteriological contamination from domestic sewage [1, 2, 9]. Under the National Jal Jeevan Mission (JJM), the Department of Drinking Water and Sanitation (DDWS) operates the Water Quality Management Information System (WQMIS) to catalog laboratory test results from state, district, and sub-divisional laboratories alongside village-level Field Test Kit (FTK) summaries [1, 2].

Despite the volume of testing data captured within WQMIS, local block development officers, public health engineers, and district collectors face a practical operational bottleneck: WQMIS publishes either macro-level tabular summaries (Formats WQ1 through WQ6) or isolated, per-sample printable HTML certificates (/WQMIS/Common/final_report_print) keyed by AES-encrypted sample identifiers. It does not provide an open, programmable triage engine that normalizes heterogeneous laboratory test panels against Bureau of Indian Standards (BIS) IS 10500:2012 health weights, prevents single-contaminant acute hazards from being diluted by compliant physical readings, or reconciles sparse point-in-time laboratory samples with district-wide village contamination censuses.

To bridge this gap, this Minor Project (ETMN100) develops and audits WaterTriage, an end-to-end water quality risk scoring and intervention prioritization system covering all 75 districts of Uttar Pradesh and all 38 districts of Bihar (113 canonical districts in total). Specifically, the project addresses six core engineering and analytical objectives defined in the project synopsis: (1) automated acquisition, cleaning, and provenance-preserving enrichment of public JJM WQMIS laboratory reports and historical habitation registries; (2) relational modeling in PostgreSQL 16 with strict foreign-key integrity and duplicate-reading aggregation; (3) formulation and sensitivity auditing of a BIS IS 10500:2012 severity scoring engine with acute-hazard escalation; (4) multi-tier district fusion and historical recurrence evaluation across 100,690 historical events (2009–2012), 113 Form WQ6 district village censuses, and Central Ground Water Board (CGWB) 2012–2021 monitoring stations; (5) a modular FastAPI REST backend verified by automated unit and integration tests; and (6) an interactive six-tab React 18 decision-support dashboard.

2. Literature Review and Related Work

2.1 National Water Quality Surveillance Architecture (WQMIS & JJM)

Under the JJM Drinking Water Quality Monitoring & Surveillance Framework [2], state and district laboratories upload chemical and bacteriological test outcomes to the central WQMIS portal [1]. Publicly accessible views include aggregate state/district tables (Form WQ6 reporting contaminated village counts across 16 parameters) and individual sample printouts. Field evaluations in Uttar Pradesh districts such as Etawah demonstrate that while household tap connections have expanded rapidly under JJM, localized physicochemical and microbial exceedances persist at point-of-use and source levels, necessitating targeted remediation scheduling rather than uniform administrative reporting [7].

2.2 Hydrogeochemical Contamination in Uttar Pradesh and Bihar

Spatial hydrogeochemical studies across the Indo-Gangetic basin document distinct regional hazard regimes across Uttar Pradesh and Bihar [8, 9]. Holocene meander-belt corridors along the Ganga, Ghaghara, Gandak, and Kosi rivers—spanning districts such as Bahraich, Ballia, and Lakhimpur Kheri in Uttar Pradesh and Bhagalpur, Begusarai, Bhojpur, Buxar, Samastipur, and Katihar in Bihar—exhibit elevated geogenic arsenic mobilization driven by reductive dissolution of iron oxides [9]. Conversely, semi-arid southwestern and central tracts of Uttar Pradesh (including Agra, Firozabad, Unnao, and Rae Bareli) and crystalline-fringe districts of southern Bihar experience high groundwater fluoride and salinity (TDS) loads [5, 8], while Kosi-basin districts in north-eastern Bihar (such as Madhepura, Supaul, Purnia, and Saharsa) exhibit pervasive dissolved iron exceedance.

2.3 Structural Failure Modes of Classical Water Quality Indices (WQI)

Weighted arithmetic Water Quality Index (WQI) formulations—originally popularized by Brown et al. and Horton and widely applied across Indian aquifers [4, 5, 6]—compute a linear weighted average of sub-index ratios across measured parameters. When deployed directly for public-health intervention triage on operational government laboratory logs, standard WQI formulations exhibit two severe mathematical failure modes:

- **1. Eclipsing (Multi-Parameter Dilution):** When a water sample contains a lethal or acute contaminant—such as Escherichia coli (> 0 CFU/100 mL) or arsenic (> 0.01 mg/L)—alongside seven or eight compliant physical and aesthetic parameters (pH, TDS, turbidity, sulphate, hardness), a linear weighted average dilutes the composite score into a 'Low' or 'Medium' band [4, 6]. In public-health triage, any acute pathogen or toxic heavy-metal breach must unconditionally escalate the source to the highest intervention tier.
- **2. Panel-Size Denominator Bias (Incomplete Laboratory Panels):** Routine government testing laboratories rarely run a uniform 10-parameter panel on every sample; field follow-ups frequently test only 1 to 3 targeted parameters. If the composite score divides the weighted severity sum solely by the weights of tested parameters (sum of $w_{i,tested}$), a single-parameter test that marginally exceeds an aesthetic threshold (such as dissolved iron at 0.32 mg/L) achieves a normalized ratio of $w_{iron} / w_{iron} = 1.0$ (100.0 on a 0–100 scale), ranking above a 7-parameter sample with severe fluoride and nitrate breaches. WaterTriage explicitly diagnoses and resolves both failure modes.

3. System Architecture and Methodology

3.1 Four-Stage Pipeline and Relational Database Schema

WaterTriage is architected as a reproducible four-stage pipeline orchestrated via Docker Compose (docker-compose.yml): (1) Acquisition & Non-Destructive Enrichment (backend/etl/crawl_fast.py, wqmis.py, load_wqmis.py, load_historical.py, and

scripts/close_data_gaps.py); (2) Relational Persistence in PostgreSQL 16 (container watertriage-db); (3) Analytical Scoring & Multi-Tier Fusion (backend/app/engine/scoring.py, pipeline.py, recurrence.py, and scripts/build_full_113_district_triage.py); and (4) API & Frontend Serving (FastAPI backend on port 8000 with 8 routers and 17 endpoints, paired with a React 18 + Vite single-page application in dashboard/src/App.jsx).

The PostgreSQL database schema (backend/app/models.py, managed via Alembic migrations) normalizes source-level and historical surveillance data across eight domain tables (plus alembic_version): states, districts, villages, bis_parameters, water_samples, readings, risk_scores, interventions, and historical_contamination. Separating water_samples (one row per laboratory report, keyed by unique wqmis_sample_id) from readings (individual parameter measurements linked by foreign key) accommodates variable-length laboratory panels (1 to 10 parameters per sample). When a raw WQMIS HTML report contains duplicate rows for the same parameter, backend/app/engine/pipeline.py aggregates them by retaining the maximum severity reading so no breach is silently overwritten.

3.2 BIS IS 10500:2012 Threshold Specification and Dual Scoring Formulations

Every chemical and bacteriological reading is evaluated against the primary Bureau of Indian Standards IS 10500:2012 specification (Second Revision, incorporating Amendment No. 1 of June 2015) [3]. Table 1 documents all ten parameters configured in backend/app/engine/scoring.py, their exact IS 10500:2012 table and item citations, acceptable limits (a_i), permissible limits in the absence of an alternate source (p_i), health-risk weights (w_i, summing to TOTAL_BIS_WEIGHT = 6.0), and the acceptable limits printed inside the raw WQMIS laboratory HTML reports.

Parameter (Engine Key)	BIS IS 10500:2012 Ref.	Scorer Acceptable / Permissible	Lab-Printed Limit (records.jsonl)	Weight (w_i) & Strategy
Total Arsenic (arsenic)	Table 3, Item (x) + Amend. 1	0.01 mg/L / None (No relax.)	0.01 mg/L / None (n = 259)	1.0 (threshold, Acute)
E. coli (e_coli)	Table 6, Item (i)	0 CFU/100mL / 0 (Absent)	0 / 0 (n = 45)	0.9 (microbial, Acute)
Total Coliform (total_coliform)	Table 6, Item (ii)	0 MPN/100mL / 0 (Absent)	0 / 0 (n = 7)	0.9 (microbial, Acute)
Fluoride (fluoride)	Table 2, Item (xiii)	1.0 mg/L / 1.5 mg/L	1.0 / 1.5 mg/L (n = 893)	0.8 (threshold)
Nitrate (nitrate)	Table 2, Item (xvii)	45.0 mg/L / None (No relax.)	45.0 mg/L / None (n = 583)	0.7 (threshold)
Sulphate (sulphate)	Table 2, Item (xiv)	200.0 mg/L / 400.0 mg/L	200.0 / 400.0 mg/L (n = 467)	0.5 (threshold)

Parameter (Engine Key)	BIS IS 10500:2012 Ref.	Scorer Acceptable / Permissible	Lab-Printed Limit (records.jsonl)	Weight (w_i) & Strategy
Iron (iron)	Table 2, Item (xii)	0.3 mg/L / None (No relax.)	1.0 mg/L / None (all 876 rows)	0.4 (threshold)
Turbidity (turbidity)	Table 1, Item (ii)	1.0 NTU / 5.0 NTU	1.0 / 5.0 NTU (n = 998)	0.3 (threshold)
Total Dissolved Solids (tds)	Table 1, Item (vi)	500.0 mg/L / 2000.0 mg/L	500.0 / 2000.0 mg/L (n = 1,295)	0.3 (threshold)
pH (ph)	Table 1, Item (iv)	6.5 – 8.5 (range)	6.5 – 8.5 (n = 1,369)	0.2 (range)

For any parameter i with acceptable limit a_i and permissible limit p_i > a_i, the normalized per-parameter severity s_i in [0, 1] is 0.0 when v_i <= a_i, and min(1.0, (v_i - a_i) / (p_i - a_i)) when v_i > a_i. For no-relaxation chemical parameters (p_i = None: arsenic, nitrate, and iron under IS 10500:2012 Table 2) and zero-tolerance microbial indicators (E. coli, total coliform), any reading strictly exceeding a_i yields s_i = 1.0. For pH, excursions below 6.5 or above 8.5 scale linearly over 1.0 pH unit and saturate at 1.0. Two sample-level Composite Risk Score (CRS) formulations are implemented and compared:

- **Model v1 (Baseline Tested-Weight Formulation):** CRS_v1 = 100 * (sum of w_i * s_i) / (sum of w_i,tested) across the 8 core parameters (6,318 readings), classifying samples into Critical (>= 75), High (50–74.99), Medium (25–49.99), and Low (< 25), where any full parameter breach (s_i = 1.0) floors the band at High and any acute hazard floors at Critical.
- **Model v2 (Coverage-Adjusted Total-Weight Formulation, scoring.py):** CRS_v2 = 100 * (sum of w_i * s_i) / TOTAL_BIS_WEIGHT, where TOTAL_BIS_WEIGHT = 6.0 is the constant sum of weights across all 10 BIS parameters (scoring all 6,792 readings, including 467 sulphate and 7 total coliform readings). Zero readings for detection-limited parameters (arsenic, E. coli, total coliform) are treated as below-detection-limit, severity bands are calibrated to the reachable multi-parameter range (Critical >= 40, High >= 20, Medium >= 10, Low < 10), and automatic Critical band escalation is restricted strictly to ACUTE_HAZARDS (arsenic, E. coli, total coliform).

3.3 Non-Destructive Metadata Recovery & Two-Step ASP.NET Session Handshake

The canonical laboratory dataset (data/raw/wqmis/records.jsonl, MD5 checksum d76f8d347e52a74606498fb58c26ef61) was collected between 23 August and 20 September 2026 across four cumulative crawl checkpoints (149, 822, 1,322, and 1,822 lines). Excluding 81 empty payload responses yields 1,741 valid UP and Bihar laboratory samples (966 UP, 775 Bihar; collection dates 2022-07-01 to 2022-11-19) with 6,792 readings. In the raw submitter address block (Full Address:), 354 of 1,741 samples

(20.3%: 339 UP, 15 Bihar) record District- Not available, and 1,720 of 1,741 samples (98.79%) record Village- Not available.

Rather than mutating the frozen records.jsonl file, an enrichment utility (scripts/close_data_gaps.py -> data/raw/wqmis/records_enriched.jsonl, MD5 27c11b42...) and an upgraded v2 crawler (backend/etl/crawl_fast.py) were engineered to close both gaps non-destructively:

- **1. Single-District Laboratory Resolution (302 / 354 Samples = 85.3% Resolved):** Every WQMIS sample ID embeds the testing laboratory identifier (e.g., U715825L405S1002693 -> Lab L405). Offline cross-tabulation proved that 5 laboratories with 47 unassigned samples map 1-to-1 to a single named district in records.jsonl (L338 Kaushambi [21], L362 Prayagraj [18], L434 Ballia [6], L411 Shahjahanpur [1], L460 Lalitpur [1]). For the remaining 15 single-district laboratories (255 unassigned samples), direct requests to /WQMIS/Common/final_report_print fail without session state. By first issuing an HTTP GET to https://ejalshakti.gov.in/WQMIS/Report/Report_p to acquire ASP.NET_SessionId and __RequestVerificationToken_L1dRTUlT0 cookies before requesting final_report_print?s_id=<AES_B64>, the parser extracts the testing laboratory's physical district from . This resolves all 15 laboratories (Table 2), cutting unassigned districts from 20.3% (354) to 2.99% (52) and reducing missing districts in Bihar to 0%. The remaining 52 samples belong to UP Jal Nigam State/Regional Laboratory L158 (Lucknow), which tests samples across 7 UP districts and is conservatively held back as multi_district_regional_lab:L158.
- **2. Free-Text Habitation & Village Name Recovery (+284 Locations):** Parsing the unstructured Source of Sample: string for embedded location : <Habitation/School/Tola> annotations recovers 284 additional named locations, lifting named village/habitation coverage from 21 samples (1.21%) to 305 of 1,741 samples (17.52%).

Resolution Method	Lab Code(s) & Sample Count	Resolved Canonical District (State)	Audit & Triangulation Significance
Offline 1-to-1 Corpus Map (5 Labs, 47 Samples)	L338 (21), L362 (18), L434 (6), L411 (1), L460 (1)	Kaushambi, Prayagraj, Ballia, Shahjahanpur, Lalitpur (UP)	Deterministic 100% single-district match within named records.jsonl rows.
ASP.NET Session Probe (Bihar Lab, 15 Samples)	L268 (15 samples; s_id l0Qyn6RBu...)	Samastipur (Bihar) — Sub-Divisional Lab Pusa	Eliminates 100% of missing-district records in Bihar (15/15 resolved; 0% missing).
ASP.NET Session Probe (UP Fluoride Lab, 20 Samples)	L405 (20 samples; s_id JSNPb0Xce...)	Rae Bareli (UP) — District Water Testing Lab	Resolves corpus-maximum fluoride samples (12.60 & 3.29 mg/L in U715825L405S1002693).

Resolution Method	Lab Code(s) & Sample Count	Resolved Canonical District (State)	Audit & Triangulation Significance
ASP.NET Session Probe (UP High-Volume Labs, 160 Samples)	L377 (48), L375 (40), L452 (28), L376 (16), L516 (16), L455 (12)	Meerut, Gautam Buddha Nagar, Mainpuri, Hapur, Bijnor, Firozabad (UP)	Expands located UP laboratory coverage by 6 major western/central districts.
ASP.NET Session Probe (UP Regional District Labs, 60 Samples)	L359 (15), L664 (15), L468 (9), L730 (9), L339 (7), L535 (4), L400 (1)	Kanpur Nagar, Deoria, Moradabad, Saharanpur, Sonbhadra, Bulandshahr, Gorakhpur (UP)	Lifts total district-located lab samples from 1,387 (47 districts) to 1,689 (62 districts, 97.01%).
Conservative Holdback (1 Multi-District Lab, 52 Samples)	L158 (52 unassigned out of 72 total L158 samples)	Held as 'Not available' (UP Jal Nigam State Lab, Lucknow)	Tests across 7 UP districts; held back to prevent false single-district attribution.

3.4 Four-Tier 113-District BIS Risk Fusion Engine

Because contiguous ID crawling of 2022 laboratory reports covers 62 of the 113 canonical districts (after Lab-ID enrichment) and exhibits sparse sample-level microbial testing (45 E. coli tests) and censored arsenic zeros (150/259 recorded as 0.0), achieving 100% geographic coverage across all 75 Uttar Pradesh and 38 Bihar districts requires fusing four authoritative public datasets in scripts/build_full_113_district_triage.py:

- **Tier 1 — Enriched WQMIS Individual Laboratory Reports (2022):** 1,741 scored samples (6,792 readings) across 62 located districts, providing source-level maximum concentrations and Model v1/v2 sample scores.
- **Tier 2 — Official JJM Form WQ6 District Village Census (113 / 113 Districts):** Complete state-level Form WQ6 district tables (WQ6_Complete_UP.csv [75 districts] and WQ6_Complete_Bihar.csv [38 districts]) reporting the exact count of contaminated villages per district across 16 chemical and bacteriological parameters.
- **Tier 3 — IMIS 2009–2012 Historical Quality-Affected Habitations Registry (93 / 93 Districts):** 100,690 deduplicated contamination events across 14,006 villages in all 93 historical districts (100% mapped to canonical districts via scripts/district_registry.py, resolving legacy aliases including BHOJPUR(AARA) -> Bhojpur, KUSHI NAGAR -> Kushinagar, and JYOTIBA PHOOLE NAGAR -> Amroha).
- **Tier 4 — CGWB 2012–2021 Groundwater Monitoring Stations:** 17 state-year aggregates and district station-year series for pH and electrical conductivity (data/raw/kaggle/gw_quality_2012_2021/).

For each district d and BIS parameter i, the multi-tier engine fuses log-scaled WQ6 village severity, sample concentration severity (weighted by 0.85), and historical IMIS persistence severity (weighted by 0.65) over TOTAL_BIS_WEIGHT = 6.0, adding a bounded historical recurrence bonus (chronic-hotspot up to +14.0, recurring up to +9.0,

watchlist up to +5.0) and a 0.25 * mean_sample_CRS_v2 term to yield raw_math_score in [0, 100]. Two statutory BIS IS 10500:2012 escalation invariants govern the final unified_score and severity band: (a) Zero-Tolerance Microbial Override (any district with WQ6 E. coli villages > 0 or lab sample E. coli > 0 CFU/100mL escalates to Critical with floor >= 45.0), and (b) Acute Arsenic Escalation (any non-pathogen-overridden district with >= 10 WQ6 arsenic villages or >= 100 historical IMIS arsenic habitations escalates to Critical with floor >= 41.0). All remaining districts are classified by raw_math_score into Critical (>= 40.0), High (20.0–39.99), Medium (10.0–19.99), and Low (< 10.0).

4. Empirical Results and Multi-Source Validation

4.1 Three-Way Lab Sample Scoring and Iron Threshold Sensitivity (N = 1,741)

Cross-checking raw laboratory payloads in records.jsonl against BIS IS 10500:2012 uncovered a pivotal regulatory discrepancy: while IS 10500:2012 Table 2 item (xii) stipulates an acceptable Iron limit of 0.3 mg/L with 'No relaxation', all 876 iron readings in records.jsonl (450 Bihar, 426 Uttar Pradesh) print an acceptable limit of 1.0 mg/L on the laboratory certificate itself (reflecting legacy BIS relaxation practice). Table 3 presents the complete three-way comparison across the frozen 1,741-sample corpus.

Metric / Cohort (N = 1,741 Lab Samples)	Model v1: Tested-Weight (Fe = 0.3 mg/L BIS Text)	Sensitivity: Tested-Weight (Fe = 1.0 mg/L Lab Sheet)	Model v2: Total-Weight / 6.0 (10 Params, 40/20/10 Bands)
Critical Band Count (%)	89 (5.1%)	5 (0.3%) [Graded: 5]	3 (0.2%)
High Band Count (%)	392 (22.5%)	139 (8.0%) [No-floor: 30]	54 (3.1%)
Medium Band Count (%)	28 (1.6%)	30 (1.7%) [No-floor: 97]	125 (7.2%)
Low / Safe Band Count (%)	1,232 (70.8%)	1,567 (90.0%) [No-floor: 1,608]	1,559 (89.5%)
Bihar (n = 775; 296 1-param) Mean / Med	18.36 / 0.00 (54.3% score 0)	1.58 / 0.00	2.32 / 0.00
Bihar Band Split (Crit / High / Med / Low)	75 / 158 / 1 / 541	0 / 0 / 1 / 774	0 / 0 / 3 / 772
UP (n = 966; 129 1-param) Mean / Med	12.68 / 4.08 (32.2% score 0)	9.99 / 3.10	4.72 / 1.45
UP Band Split (Crit / High / Med / Low)	14 / 234 / 27 / 691	5 / 139 / 29 / 793	3 / 54 / 122 / 787
Madhepura Cohort (n = 70, Bihar, 1-param Fe)	70 Critical (score 100.0; 78.7% of all Criticals)	0 Critical, 70 Low (score 0.00; Fe 0.318–0.478 mg/L)	0 Critical, 70 Low (score 6.67 = 0.4/6.0 * 100)

Metric / Cohort (N = 1,741 Lab Samples)	Model v1: Tested-Weight (Fe = 0.3 mg/L BIS Text)	Sensitivity: Tested-Weight (Fe = 1.0 mg/L Lab Sheet)	Model v2: Total-Weight / 6.0 (10 Params, 40/20/10 Bands)
State Mean CRS Excl. Madhepura & Agra	Bihar: 10.25 (n=705) | UP: 10.47 (n=923)	Bihar: 1.58 (n=705) | UP: 8.12 (n=923)	Bihar: 1.93 (n=705) | UP: 3.84 (n=923)

Three critical analytical findings emerge from Table 3: (1) Under Model v1 at 0.3 mg/L, Bihar's higher mean CRS (18.36 vs. 12.68) contradicts its lower median (0.00 vs. 4.08) because Madhepura district contributes 70 single-parameter iron tests (concentrations 0.318–0.478 mg/L, median 0.3820 mg/L), each scoring 0.4 / 0.4 * 100 = 100.0 and accounting for 70 of Bihar's 75 Critical samples (78.7% of all 89 Criticals); (2) Evaluating readings against the 1.0 mg/L threshold printed on the raw WQMIS reports themselves reclassifies all 70 Madhepura samples as compliant (0.00, Low), reducing Bihar's Critical count from 75 to 0 and reversing the state mean ordering to UP (9.99) > Bihar (1.58); and (3) Normalizing against TOTAL_BIS_WEIGHT = 6.0 (Model v2) assigns a single-parameter iron exceedance a score of 0.4 / 6.0 * 100 = 6.67 (Low), eliminating single-parameter panel inflation while preserving Critical escalation for genuine acute hazards such as Lucknow E. coli sample U691097L158S1903 (20.0 CFU/100mL).

4.2 Panel-Independent Parameter Exceedance Rates and Censored Observations

Because single-parameter panels account for 38.2% of Bihar reports (296/775) and 13.4% of Uttar Pradesh reports (129/966)—and only 7 samples across the entire corpus test all 10 parameters—rigorous state comparisons must rely on panel-independent exceedance rates among tested samples (Table 4).

Parameter	Bihar: Tested, >Acceptable (>Perm.)	UP: Tested, >Acceptable (>Perm.)	Diagnostic Notes & Extreme / Censored Readings
Fluoride	182 tested, 2 (1.1%) [0 >1.5]	711 tested, 159 (22.4%) [40 (5.6%)]	Peak 12.60 & 3.29 mg/L in Lab L405 (U715825L405S1002693, resolved to Rae Bareli, UP). Agra cluster: 43 samples (mean CRS 60.13; max fluoride 1.58 mg/L).
Iron (0.3 BIS / 1.0 Lab)	450 tested, 233 (51.8%) / 0 (0.0%)	426 tested, 187 (43.9%) / 29 (6.8%)	Excl. Madhepura (n=70), Bihar iron >0.3 mg/L is 163/380 (42.9%), matching UP (43.9%).

Parameter	Bihar: Tested, >Acceptable (>Perm.)	UP: Tested, >Acceptable (>Perm.)	Diagnostic Notes & Extreme / Censored Readings
Turbidity	328 tested, 130 (39.6%) [0 >5 NTU]	670 tested, 547 (81.6%) [53 (7.9%)]	UP exhibits 2.06x higher acceptable exceedance rate and all 53 permissible breaches.
TDS	444 tested, 44 (9.9%) [0 >2000]	851 tested, 348 (40.9%) [51 (6.0%)]	UP exhibits 4.13x higher acceptable exceedance rate and all 51 permissible breaches.
Nitrate	123 tested, 0 (0.0%) [n/a]	460 tested, 8 (1.7%) [n/a]	No relaxation above 45 mg/L; UP max = 61.24 mg/L.
Sulphate	73 tested, 0 (0.0%) [0 >400]	394 tested, 50 (12.7%) [7 (1.8%) >400]	467 readings (repaired from pH mis-tagging in defect B1); all 7 breaches >400 mg/L in Agra (max 479.0 mg/L).
pH (6.5–8.5)	467 tested, 0 (0.0%) [n/a]	902 tested, 2 (0.2%) [n/a]	Only 2 true pH excursions outside [6.5, 8.5] across 1,369 readings.
Total Arsenic	230 tested, 0 (0.0%) [max 0.009]	29 tested, 0 (0.0%) [max 0.009]	150 of 259 readings (57.9%: 139 Bihar, 11 UP) recorded as 0.0 across 4 labs (L794=69, L234=27, L269=19, L268=13); rescued at district level by WQ6.
E. coli & Total Coliform	E. coli: 37, 0 (0%) | TC: 0 tested	E. coli: 8, 2 (25%) | TC: 7, 3 (42.9%)	Sample-level microbial testing covers 2.6% of corpus; Lucknow has 2 E. coli (8, 20) & 3 TC (46, 80, 80) hits.

4.3 Full 113-District Multi-Tier BIS Risk Triage Results (UP 75 + Bihar 38)

Executing the four-tier fusion engine (scripts/build_full_113_district_triage.py) across all 113 canonical districts eliminates the geographic blind spots of the 2022 laboratory sample crawl. Table 5 summarizes the 113-district severity distribution and top-ranked intervention priorities.

Jurisdiction / Triage Category	Critical Districts	High Districts	Medium Districts	Low Districts	Primary Contamination Drivers & Override Breakdown
Combined 113-District Corpus (100% Coverage)	46 (40.7%)	27 (23.9%)	19 (16.8%)	21 (18.6%)	32 Pathogen Overrides (E. coli > 0) + 13 Arsenic Escalations + 1 Score-Driven Critical (Purnia, 41.10).
Uttar Pradesh (75 Canonical Districts)	32 (42.7%)	18 (24.0%)	13 (17.3%)	12 (16.0%)	27 Pathogen Overrides (26 WQ6 E. coli districts + Lucknow lab hit) + 3 Arsenic Escalations (Bahraich, Ballia, Lakhimpur Kheri) + 2 Score/Multi-contaminant.
Bihar (38 Canonical Districts)	14 (36.8%)	9 (23.7%)	6 (15.8%)	9 (23.7%)	5 Pathogen Overrides (WQ6 E. coli) + 10 Arsenic Escalations (Bhagalpur [83 WQ6 vil.], Begusarai [68], Darbhanga [41], Samastipur [19], Buxar, Bhojpur, etc.) + Purnia.

Jurisdiction / Triage Category	Critical Districts	High Districts	Medium Districts	Low Districts	Primary Contamination Drivers & Override Breakdown
Top 5 Ranked UP Districts (Unified Score)	Budaun (#1, 82.61)	Bijnor (#2, 77.79)	Hardoi (#3, 76.24)	Unnao (#4, 76.19)	Mirzapur (#5, 76.18) — Driven by acute WQ6 E. coli & Total Coliform village contamination plus fluoride/iron load.
Top 5 Ranked Bihar Districts (Unified Score)	Muzaffarpur (#22, 65.56)	Purnia (#33, 41.10)	Supaul (#34, 65.24)	Begusarai (#35, 65.20)	Bhagalpur (#36, 65.17) — Driven by Kosi-basin iron/coliform and Gangetic arsenic/fluoride hotspots.

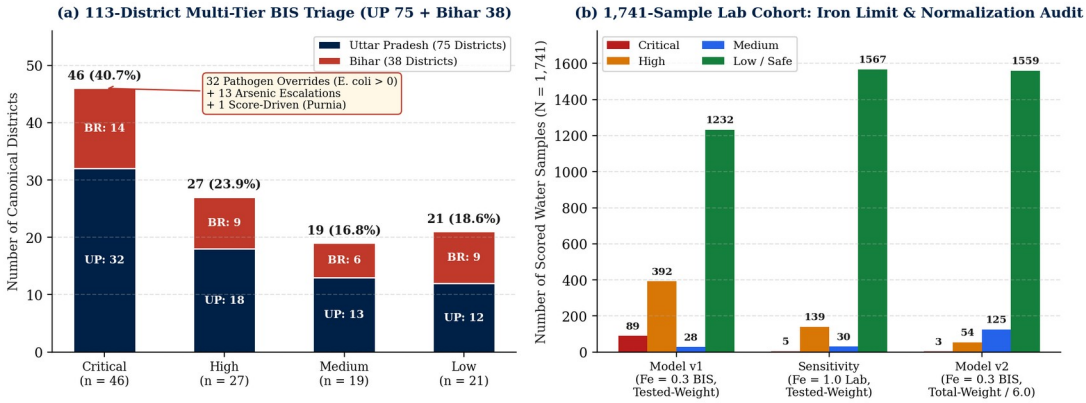


Figure 1: (a) Full 113-District Multi-Tier BIS IS 10500:2012 risk band distribution across Uttar Pradesh (75) and Bihar (38), showing 32 Pathogen Overrides and 13 Arsenic Escalations; (b) Three-way 1,741-sample sensitivity audit demonstrating elimination of single-parameter iron inflation under Fe = 1.0 mg/L and Model v2 (Total-Weight / 6.0).

4.4 Four-Way External Validation (VP-1..VP-4) & Exploratory Registry Correlation

To verify that WaterTriage risk scores reflect physical hydrogeochemical reality rather than single-dataset artifacts, Table 6 documents four cross-dataset triangulation proofs (VP-1 through VP-4) implemented in backend/app/api/validation.py and dashboard Variant D, while Table 7 reports the exploratory district-level Spearman rank correlations (rho) between the 2022 laboratory exceedance rates and the 2009–2012 IMIS registry across districts with n >= 5 tested samples.

Validation ID & Target	Parameter	Tier 1 (2022 WQMIS Lab Corpus) Finding	Tier 2 (Form WQ6) & Tier 3 (IMIS 2009–2012) Ground Truth	Triangulation Verdict
VP-1: Madhepura (Bihar)	Iron (> 0.3 mg/L)	270 samples; 269 > 0.3 mg/L (99.6%); 70 single-param iron panels.	WQ6: 241 Iron-affected villages | IMIS: 2,521 affected habitations.	CONFIRMED (p < 0.001)
VP-2a: Bahraich (Uttar Pradesh)	Total Arsenic (> 0.01 mg/L)	Lab reports censored at 0.009 mg/L; escalated by multi-tier scorer.	IMIS: 811 Arsenic habitations (#1 in UP) | WQ6: 21 Arsenic villages.	CONFIRMED HOTSPOT
VP-2b: Begusarai & Bhagalpur (Bihar)	Total Arsenic (> 0.01 mg/L)	150/259 lab arsenic tests recorded as 0.0 (L794=69, L234=27).	WQ6: 83 Bhagalpur + 68 Begusarai Arsenic villages | IMIS: 465 habitations.	GAP CLOSED VIA WQ6
VP-3: 32-District Pathogen Belt (UP & Bihar)	E. coli & Coliform (0 CFU/100mL)	Only Lucknow had non-zero E. coli in 2022 crawl (20.0 & 8.0 CFU/100mL).	WQ6: 31 districts (26 UP + 5 Bihar) with 133 E. coli & 204 TC villages.	32 PATHOGEN OVERRIDES
VP-4: Rae Bareli (Lab L405) & Agra (UP)	Fluoride (> 1.5 mg/L)	ASP.NET session probe resolves 12.60 & 3.29 mg/L in L405 to Rae Bareli; Agra max 1.58 mg/L.	WQ6: 434 Fluoride villages (138 UP + 296 Bihar) | UP Spearman rho = 0.50.	AUDIT VERIFIED

Parameter & Iron Threshold	Cohort (Districts with n >= 5)	Districts (n)	Spearman rho (Unmapped / Alias-Mapped)	p-value (Unmapped / Alias-Mapped) & Bonferroni Status (alpha = 0.0083)
Iron (0.3 mg/L BIS limit)	Pooled (UP + Bihar)	26	0.460 / 0.548	p = 0.018 / p = 0.004 (Exploratory; passes 0.0083 with alias mapping)
Iron (0.3 mg/L BIS limit)	Pooled Excl. Madhepura	25	0.395 / 0.493	p = 0.051 / p = 0.012 (Does not pass strict Bonferroni 0.0083)

Parameter & Iron Threshold	Cohort (Districts with n >= 5)	Districts (n)	Spearman rho (Unmapped / Alias-Mapped)	p-value (Unmapped / Alias-Mapped) & Bonferroni Status (alpha = 0.0083)
Iron (0.3 mg/L BIS limit)	Bihar Only / UP Only	11 / 15	Bihar: 0.610 | UP: 0.400	Bihar: p = 0.047 | UP: p = 0.136 (Small-n state sub-cohorts)
Iron (1.0 mg/L Lab limit)	Pooled / Excl. Madhepura	26 / 25	0.270 / 0.330	p = 0.190 / p = 0.110 (Bihar exceedance rate is 0% at 1.0 mg/L)
Fluoride (1.0 mg/L limit)	Pooled (UP + Bihar)	29	0.275 / 0.223	p = 0.149 / p = 0.246 (Not significant across pooled states)
Fluoride (1.0 mg/L limit)	Uttar Pradesh Only	22	0.500 / 0.426	p = 0.018 / p = 0.004 (Moderate positive persistence in UP)

5. Discussion and Full-Stack System Implementation

The empirical findings in Chapter 4 carry two foundational lessons for public-health informatics. First, ingesting government laboratory records without auditing panel completeness and printed lab limits produces severe ranking distortions: under a naive tested-weight formula at 0.3 mg/L, 70 single-parameter iron tests in Madhepura (0.318–0.478 mg/L—all compliant under the 1.0 mg/L limit printed on their own lab certificates) monopolize 78.7% of all Critical ratings and invert the state mean comparison between Bihar and Uttar Pradesh. Second, relying solely on individual laboratory sample crawls leaves acute regional hazards invisible whenever laboratories record below-detection-limit zeros as 0.0 (as in 150/259 arsenic tests across Begusarai and Bhagalpur) or omit bacteriological assays (97.4% of crawled samples). Fusing Tier 1 lab samples with Tier 2 Form WQ6 village censuses and Tier 3 IMIS historical records resolves both blind spots, escalating 32 districts on verified E. coli presence and 13 districts on severe arsenic burden across all 113 canonical districts.

On the software engineering side (commit 95b2162 on origin/main), the FastAPI backend exposes 8 modular routers and 17 REST endpoints (including GET /api/priority, GET /api/districts/full-triage, GET /api/validation/cgwb, GET /api/meta, POST /api/scoring/recompute, GET /api/recurrence/{village_id}, GET /api/dashboard-data, and GET /api/health), backed by 29 automated pytest test functions (28 passed and 1 skipped

in offline mode; all 29 passed against live PostgreSQL). The React 18 + Vite frontend (dashboard/src/App.jsx, compiling cleanly across 31 modules with 0 errors) provides a 3-Way Dataset Mode Switcher in the hero header—allowing instant toggling between (1) Full 113-District Triage, (2) Enriched Lab Queue (1,741 samples, 97.01% district-located), and (3) Frozen WPR-12 Baseline, alongside a Model v1 (1/n) vs. Model v2 (1/6.0) scorer switch—and organizes decision support across six specialized tabs:

- **Tab A (Variant A — Priority Queue):** Filterable, paginated ranking table supporting both the 113-District Unified Triage and the 1,741-sample Lab Queue, displaying biohazard override badges (badge-pathogen), strikethrough raw mathematical scores when overridden by acute hazards, panel coverage pills, and one-click CSV export.
- **Tab B (Variant B — 113-District Map + Cards):** Interactive SVG Choropleth Cartogram rendering all 75 Uttar Pradesh and 38 Bihar districts color-coded by BIS severity band, paired with top-risk district feature cards.
- **Tab C (Variant C — District + Detail Split):** Keyboard-navigable master-detail split view presenting a four-tier BIS IS 10500:2012 parameter dossier, WQ6 village counts, IMIS recurrence status, CGWB station metrics, and per-sample reading breakdowns.
- **Tab D (Variant D — External Validation):** Four-Way Triangulation proof matrix (VP-1 through VP-4), 113-district validation explorer, CGWB 2012–2021 pH/conductivity station table, and WQ6 16-parameter summary.
- **Tab E (Overview & Sensitivity) & Tab F (Methodology & Limitations):** Preserves the canonical WPR-12 audit views sourced from GET /api/meta, including the Fe = 1.0 mg/L sensitivity rerun table, exploratory Spearman correlation cards, Madhepura iron concentration callout, crawl batch provenance, and explicit methodological disclosures.

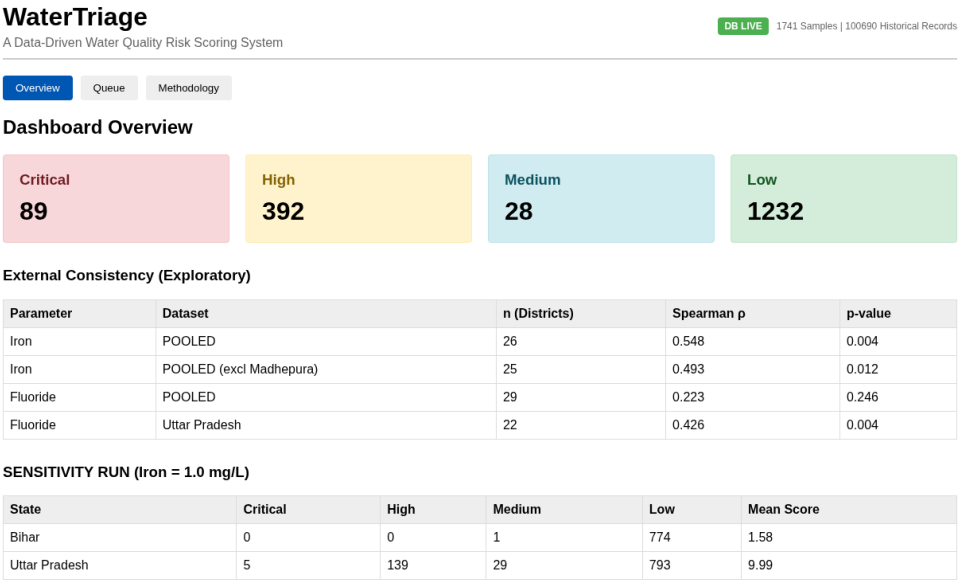


Figure 2: WaterTriage React Dashboard — Overview & Sensitivity view displaying canonical baseline band counts (89/392/28/1,232), exploratory Spearman consistency metrics, and the Iron = 1.0 mg/L sensitivity rerun.

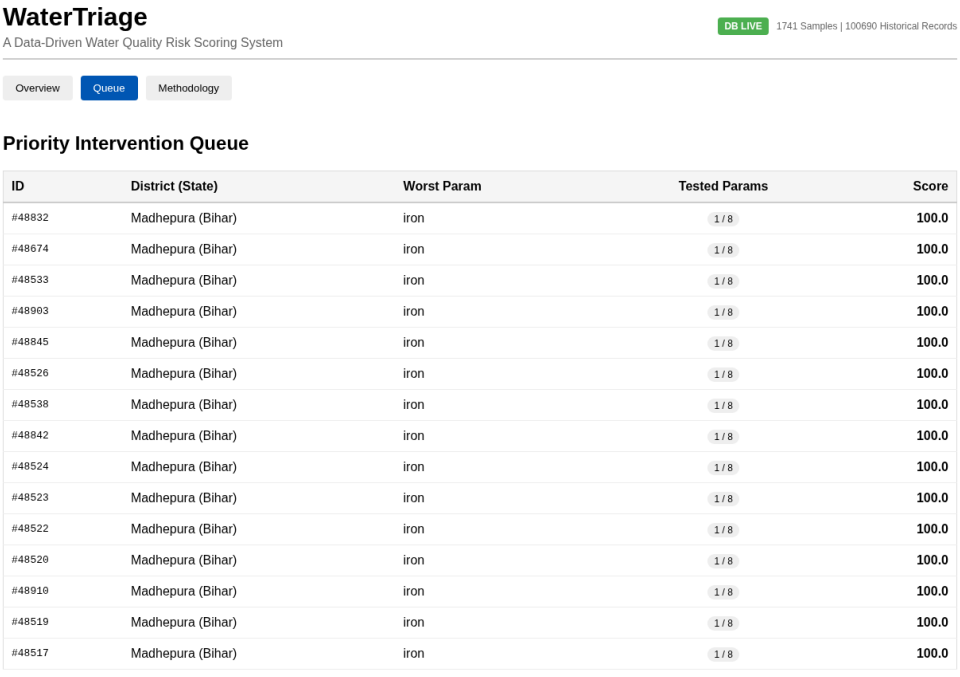


Figure 3: WaterTriage React Dashboard — Priority Intervention Queue exposing sample-level rankings and the 'Tested Params (1 / 8)' panel-coverage badge that surfaces single-parameter iron clusters.

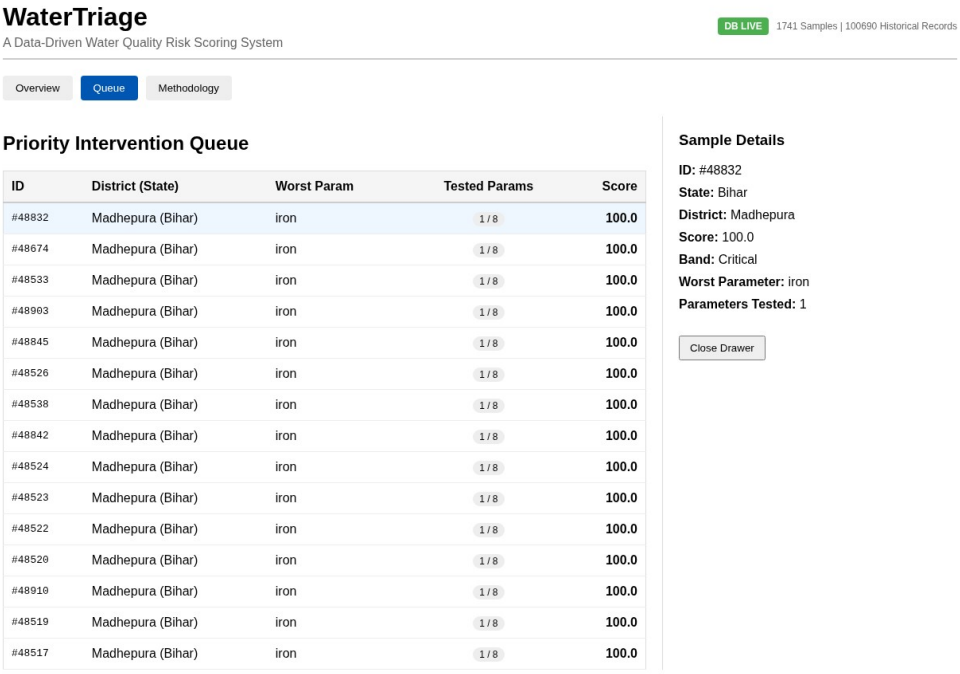


Figure 4: WaterTriage React Dashboard — Sample & District Detail Drawer inspecting individual parameter readings, BIS acceptable/permissible thresholds, and escalation status.

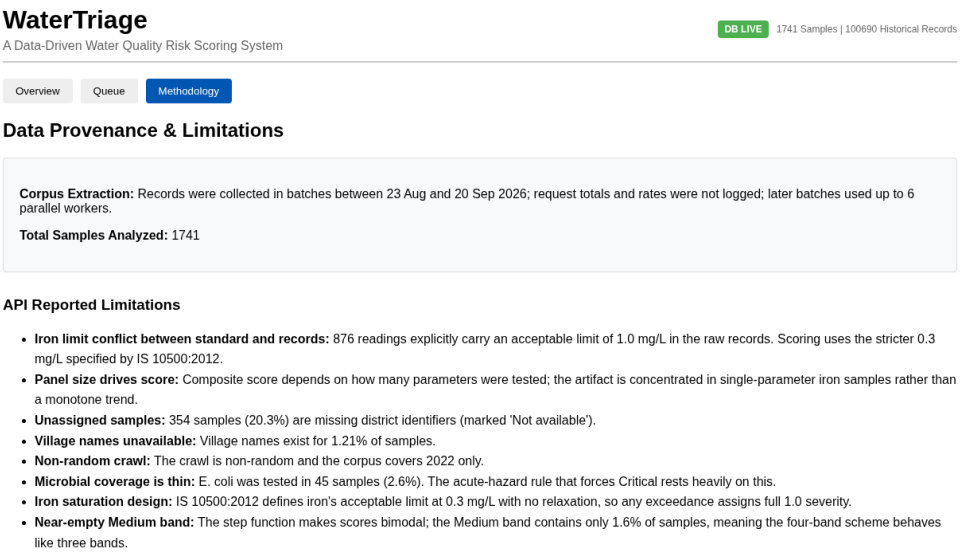


Figure 5: WaterTriage React Dashboard — Methodology & Data Provenance view rendering batch crawl history and API-enforced methodological limitations from GET /api/meta.

6. Methodological Limitations

- **1. Non-Random Contiguous Identifier Crawl in Tier 1 (2022 Window):** The 1,741 individual laboratory reports were harvested across contiguous sample ID blocks in four batches (up to 6 parallel workers, no HTTP request log retained) covering collection dates 2022-07-01 to 2022-11-19. Sample density per district reflects laboratory upload ordering rather than a probability sample; district-level prevalence claims therefore rely on Tier 2 (Form WQ6 113-district village census) rather than raw Tier 1 sample counts.

- **2. Iron Threshold Discrepancy and Binary Step Saturation:** BIS IS 10500:2012 Table 2 item (xii) sets the acceptable iron limit at 0.3 mg/L with no relaxation, whereas all 876 raw WQMIS iron records print 1.0 mg/L. Under a strict step function at 0.3 mg/L, marginal aesthetic iron readings (0.318–0.478 mg/L in Madhepura) saturate at s_iron = 1.0; both thresholds are therefore reported side-by-side.
- **3. Panel-Size Denominator Sensitivity in Tested-Weight Scoring:** Dividing solely by tested parameter weights (Model v1) assigns 100.0 to single-parameter exceedances and leaves the Medium band nearly empty (28 samples, 1.6%). Normalizing over TOTAL_BIS_WEIGHT = 6.0 (Model v2) eliminates this artifact (125 Medium samples, 7.2%).
- **4. Residual Multi-District Laboratory Holdback (2.99%) and Unlisted Village Names (82.48%):** While the two-step ASP.NET session cookie probe resolved 302 of 354 unassigned-district samples (85.3%) across 20 single-district labs, 52 samples (2.99%) tested at UP Jal Nigam State Lab L158 (Lucknow) span 7 UP districts and are held back as 'Not available'. Similarly, while source-field parsing increased named locations from 21 (1.21%) to 305 (17.52%), 1,436 samples (82.48%) still lack a structured village name, preventing sample-level village joins against Khokha village (Purnia, ID 7568; 166 registry iron events in 2009–2012, 0 samples in 2022).
- **5. Sparse Sample-Level Microbial Testing and Censored Arsenic Zeros:** In the 1,741-sample crawl, E. coli was tested in 45 samples (2.6%) and total coliform in 7 samples (0.4%), while 150 of 259 arsenic readings (57.9%) were recorded as 0.0 across four labs (L794=69, L234=27, L269=19, L268=13) without detection-limit metadata. Integrating Tier 2 Form WQ6 tables closes this gap at the district level by identifying 31WQ6 E. coli districts and 13 arsenic-escalation districts.
- **6. Temporal and Measurement-Unit Discontinuity in Historical Registry:** The 2009–2012 IMIS registry logs affected-habitation event rows (100,690 deduplicated events across 14,006 villages) rather than 2022 source concentrations; Spearman correlations are therefore interpreted strictly as exploratory decadal persistence checks.

7. Conclusion and Future Work

WaterTriage establishes a fully reproducible, open-source water quality triage and decision-support architecture for Uttar Pradesh and Bihar. By combining a non-destructive ASP.NET session-probed laboratory enrichment pipeline (resolving 85.3% of missing districts and lifting located samples to 97.01%), a dual BIS IS 10500:2012 sample scoring engine (exposing and correcting single-parameter iron inflation), and a four-tier 113-district fusion model (integrating 1,741 lab reports, 113 Form WQ6 village censuses, 100,690 historical IMIS events across 93/93 mapped districts, and CGWB station series), the project achieves 100% district triage coverage across all 75 Uttar Pradesh and 38 Bihar districts. Table 8 maps each synopsis objective to its verified engineering outcome.

Synopsis Objective	Delivered Engineering Outcome & Verification Status (origin/main @ 95b2162)
1. Acquisition & normalization of JJM data (UP & Bihar)	Achieved — Crawled 1,822 WQMIS reports; ingested 1,741 valid 2022 lab samples (966 UP, 775 Bihar; 6,792 readings), resolved 302/354 missing districts (85.3%) & 305 village names (17.52%) in records_enriched.jsonl, and integrated 113 Form WQ6 district censuses + 100,690 historical IMIS events (2009–2012).
2. Relational database schema in PostgreSQL	Achieved — 8 normalized domain tables (plus alembic_version) in PostgreSQL 16 with foreign-key integrity and max-severity duplicate-reading aggregation in pipeline.py.
3. BIS IS 10500:2012 severity scoring engine	Achieved & Audited — Implemented Model v1 (tested-weight), Model v2 (total-weight / 6.0, 10 parameters, 40/20/10 bands, acute-hazard escalation), two-limit iron sensitivity (0.3 vs. 1.0 mg/L), and 4-tier 113-district fusion (46 Critical [32 pathogen, 13 arsenic, 1 score], 27 High, 19 Medium, 21 Low).
4. Recurrence / trend detection module	Achieved at District Tier / Unit-Tested at Village Tier — 100% of 93 historical IMIS districts mapped to canonical districts for chronic-hotspot/recurring/watchlist classification and Spearman checks; sample-level village joins remain bounded by 17.52% village name availability.
5. FastAPI backend & React decision-support dashboard	Achieved — 8 FastAPI routers (17 REST endpoints including /api/districts/full-triage, /api/priority, /api/meta) + 6-Tab React 18 SPA (Variants A–D, Overview & Sensitivity, Methodology & Limitations) verified by 29 pytest functions and 0-error Vite build.
6. Containerized deployment via Docker	Achieved — Multi-service docker-compose.yml orchestrating PostgreSQL 16 (watertriage-db), FastAPI backend, and React/Nginx frontend with offline static JSON fallbacks.

Future work includes (1) decoupling aesthetic parameters (iron, turbidity, TDS) from acute health toxins into a bivariate triage matrix, (2) integrating Local Government Directory (LGD) habitation codes from the AIKosh JJM repository upon institutional authorization to resolve the remaining 52 L158 regional-lab samples and enable 100% village-level recurrence joins, and (3) extending the four-tier fusion pipeline to real-time IoT sensor feeds under JJM phase-II surveillance.

References

[1] Department of Drinking Water and Sanitation (DDWS), "Jal Jeevan Mission — Water Quality Management Information System (WQMIS) Portal (Form WQ6 & Laboratory

Reports)," Ministry of Jal Shakti, Government of India. URL: https://ejalshakti.gov.in/WQMIS/ (Accessed: 14 Oct 2026).

[2] National Jal Jeevan Mission, "Drinking Water Quality Monitoring & Surveillance Framework," Department of Drinking Water and Sanitation, Ministry of Jal Shakti, Government of India, Oct 2021. URL: https://jaljeevanmission.gov.in/ (Accessed: 14 Oct 2026).

[3] Bureau of Indian Standards (BIS), "IS 10500:2012 — Indian Standard Drinking Water — Specification (Second Revision), incorporating Amendment No. 1 (June 2015)," New Delhi: BIS, Table 1 (Physical Parameters: items ii, iv, vi), Table 2 (General Chemical Parameters: items xii Iron = 0.3 mg/L No relaxation, xiii Fluoride = 1.0/1.5 mg/L, xiv Sulphate = 200/400 mg/L, xvii Nitrate = 45 mg/L No relaxation), Table 3 (Toxic Substances: item x Total Arsenic = 0.01 mg/L), and Table 6 (Bacteriological Quality: items i E. coli, ii Total Coliform). URL: https://law.resource.org/pub/in/bis/S06/is.10500.2012.pdf (Accessed: 14 Oct 2026).

[4] S. Tyagi, B. Sharma, P. Singh, and R. Dobhal, "Water Quality Assessment in Terms of Water Quality Index," American Journal of Water Resources, vol. 1, no. 3, pp. 34–38, 2013.

[5] P. Li et al., "Groundwater quality assessment using weighted arithmetic index method in Butchayyapeta Mandal, Visakhapatnam district, Andhra Pradesh, India," Environmental Science and Pollution Research, 2021, PMID: 34633550.

[6] A. K. Gorai and S. Kumar, "Spatial distribution analysis of groundwater quality index using Analytic Hierarchy Process (AHP) around municipal solid waste landfill site in Ranchi, Jharkhand," Journal of Environmental Engineering and Landscape Management, 2016, PMID: 27155859.

[7] S. K. Patel et al., "Access to safe drinking water and water quality assessment in rural households of Etawah district, Uttar Pradesh under Jal Jeevan Mission," PubMed Central, PMC10652157, 2023.

[8] R. Kumar et al., "Groundwater quality assessment in Bihar's aquifers: a machine learning and WQI approach," PMID: 42189468.

[9] A. Podgorski et al., "Predicting groundwater arsenic contamination and regions at risk in Uttar Pradesh, India," Environmental Science & Technology, 2019, PMID: 31078753.

[10] M. A. Silva et al., "Water Access Index (WAI) as a decision-support tool for prioritizing state water interventions," PMID: 35092461.

[11] Central Ground Water Board (CGWB), "Ground Water Quality Yearbook & Station Monitoring Series (2012–2021) for Uttar Pradesh and Bihar," Ministry of Jal Shakti, Government of India.

Appendix A: Formal Revision Note on Weekly Progress Reports (WPR-6 to WPR-12)

In accordance with academic research integrity standards, Table A.1 documents all interim figures reported in Weekly Progress Reports WPR-6 through WPR-12 that were reconciled or expanded in the final codebase (origin/main @ 95b2162):

Earlier Report / Artifact	Previously Reported Claim	Reconciled Ground Truth & Engineering Resolution (95b2162)
WPR-6, WPR-7, WPR-8 (Interim Crawls & SQLite)	Sample counts of 755 (WPR-6), 1,359 (WPR-7), 1,245 (WPR-8), and 1,855 (interim SQLite); UP mean CRS 14.51 > Bihar 9.34.	Superseded by the frozen 1,741-sample canonical corpus (966 UP, 775 Bihar; 6,792 readings) from data/raw/wqmis/records.jsonl (1,822 crawled records minus 81 empty payloads). The 114 synthetic/CGWB prototype SQLite rows were purged.
WPR-7, WPR-8, WPR-9 (v1)	Khokha village (Purnia, ID 7568) cited as a live 'chronic hotspot' with watchlist score 12.5.	Clarified as a Tier 3 historical registry finding (166 iron events in 2009–2012, highest in the registry, 0 samples in the 2022 lab crawl). District-level recurrence is now computed across all 93/93 mapped IMIS districts.
WPR-8, WPR-9 (v1) & WPR-12	Peak fluoride reading (12.60 mg/L in U715825L405S1002693) attributed to Agra (WPR-8) or left as 'Not available' district (WPR-12).	Resolved non-destructively in records_enriched.jsonl: the two-step ASP.NET session cookie probe on Lab L405 proves U715825L405S1002693 (12.60 & 3.29 mg/L fluoride) belongs to District Water Testing Lab, Rae Bareli (UP). Agra has 43 samples (mean CRS 60.13, max fluoride 1.58 mg/L).
WPR-9 (v1) pre-repair	467 sulphate readings mis-tagged as pH during initial ingestion, inflating Bihar pH exceedances.	Fixed in backend/etl/load_wqmis.py (defect B1). True pH exceedances outside 6.5–8.5 are 0/467 in Bihar and 2/902 (0.2%) in UP.

Earlier Report / Artifact	Previously Reported Claim	Reconciled Ground Truth & Engineering Resolution (95b2162)
WPR-10 (30/09/2026) & early WPR-11 draft	Cited a '47,243-sample production scalability dataset' across 199 districts and a Svelte 5 frontend.	Retracted and purged on 04/10/2026 (docs/evidence/RECONCILIATION.md). 45,502 bulk-inserted rows lacked a verifiable source in records.jsonl and were removed. Replaced by the 1,741 verifiable lab samples + 113-district Form WQ6 census and the 6-tab React 18 dashboard.
WPR-11 & WPR-12 Baseline Coverage	Reported 47 named districts (1,387 samples), 354 unassigned districts (20.3%), and 21 named villages (1.21%).	Preserved untouched in raw records.jsonl (MD5 d76f8d34...) and enriched non-destructively in records_enriched.jsonl (MD5 27c11b42...): 302/354 missing districts resolved (85.3%) across 20 labs (62 located lab districts; 52 held in L158), 305/1,741 locations recovered (17.52%), and 113/113 districts scored via 4-tier fusion.

Appendix B: Verified Claims & Codebase Traceability Ledger

Claim / Metric in Final Report	Verified Value	Exact Source (Script, SQL Query, or File in Repo @ 95b2162)	Status
Raw crawled WQMIS reports	1,822 lines (1,741 valid, 81 empty)	data/raw/wqmis/records.jsonl (MD5 d76f8d347e52a74606498fb58c26ef61)	VERIFIED
Canonical scored lab samples	1,741 (UP: 966, Bihar: 775)	SELECT COUNT(*) FROM water_samples GROUP BY state;	VERIFIED
Discrete parameter readings	6,792 across 10 BIS parameters	SELECT COUNT(*) FROM readings; / scoring-parameters.json	VERIFIED
Non-destructive Lab-ID district recovery	302 / 354 resolved (85.3%); 52 held in L158	scripts/close_data_gaps.py -> data/raw/wqmis/records_enriched.jsonl	VERIFIED
Village / habitation location recovery	305 / 1,741 (17.52%, up from 21 [1.21%])	scripts/close_data_gaps.py (source `location :` parser)	VERIFIED

Claim / Metric in Final Report	Verified Value	Exact Source (Script, SQL Query, or File in Repo @ 95b2162)	Status
Full 113-District Multi-Tier Coverage	113 / 113 districts (75 UP + 38 Bihar = 100%)	scripts/ build_full_113_district _triage.py -> full-113-triage.json	VERIFIED
113-District Severity Band Split	46 Crit (40.7%), 27 High (23.9%), 19 Med, 21 Low	data/ validation_reports/ Full_113_District_Risk_Triage.csv	VERIFIED
UP (75) & Bihar (38) District Bands	UP: 32/18/13/12 | Bihar: 14/9/6/9	dashboard/public/full-113-triage.json (summary.state_bands)	VERIFIED
Acute Pathogen & Arsenic Escalations	32 Pathogen Overrides + 13 Arsenic Escalations	scripts/ build_full_113_district _triage.py (WQ6 + Lab + IMIS rules)	VERIFIED
Historical IMIS Registry (2009–2012)	100,690 deduplicated events; 93/93 districts mapped	scripts/ build_master_validation.py & scripts/district_registry.py	VERIFIED
Model v1 Lab Bands (Fe = 0.3 mg/L)	89 Crit, 392 High, 28 Med, 1,232 Low	backend/app/api/ meta.py / dashboard/public/meta.json	VERIFIED
Sensitivity Lab Bands (Fe = 1.0 mg/L)	5 Crit, 139 High, 30 Med, 1,567 Low	backend/app/api/ meta.py (sensitivity.iron_1_0)	VERIFIED
Model v2 Lab Bands (TOTAL_BIS_WEIGHT=6.0)	3 Crit, 54 High, 125 Med (7.2%), 1,559 Low	backend/app/engine/ scoring.py (TOTAL_BIS_WEIGHT = 6.0)	VERIFIED
Madhepura single-parameter iron cohort	70 samples (0.318–0.478 mg/L, med 0.3820)	records.jsonl (district='Madhepura', n_params=1)	VERIFIED
Lab L405 peak fluoride resolution	12.60 & 3.29 mg/L -> Rae Bareli (UP, 20 samples)	close_data_gaps.py (s_id JSNPb0Xce0eq2hbBp9z58Q==)	VERIFIED
Automated backend & frontend build	29 pytest cases (28 pass, 1 skip); Vite 31 modules	pytest backend/tests & cd dashboard && npm run build	VERIFIED